

THE 2026 CONFERENCE ON ARTIFICIAL LIFE (ALIFE)

WATERLOO SPECIAL SESSION: ARTIFICIAL LIFE FOR SCIENCE & ENGINEERING

Multi-Behavioral Evolved Substrates Through Neuromodulation and Activation Selection

One evolved genotype, many phenotypic behaviors

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github.com/RomainClaret/emr-hyperneat

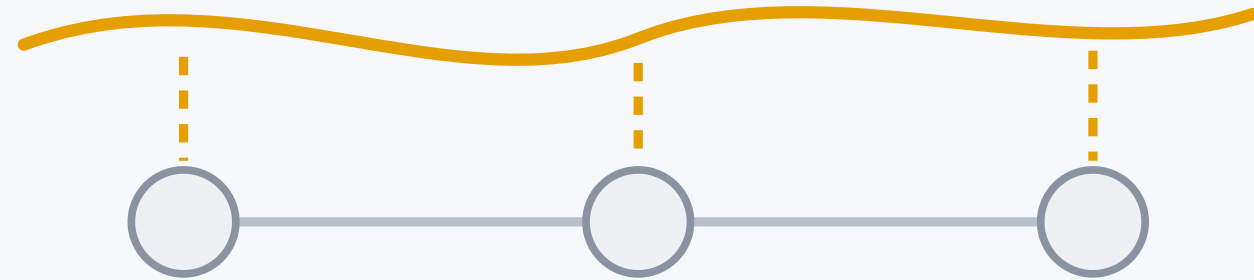


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Brains combine two mechanisms. We usually copy one.

Neuromodulation



Chemical signals **reconfigure circuits** on the fly (gain, thresholds, gating) without changing a connection.

long studied: Soltoggio → Backpropamine gating is its mainstream cousin

Diverse neuron types



Distinct input–output profiles, **matched to circuit roles** : different computational primitives in one circuit.

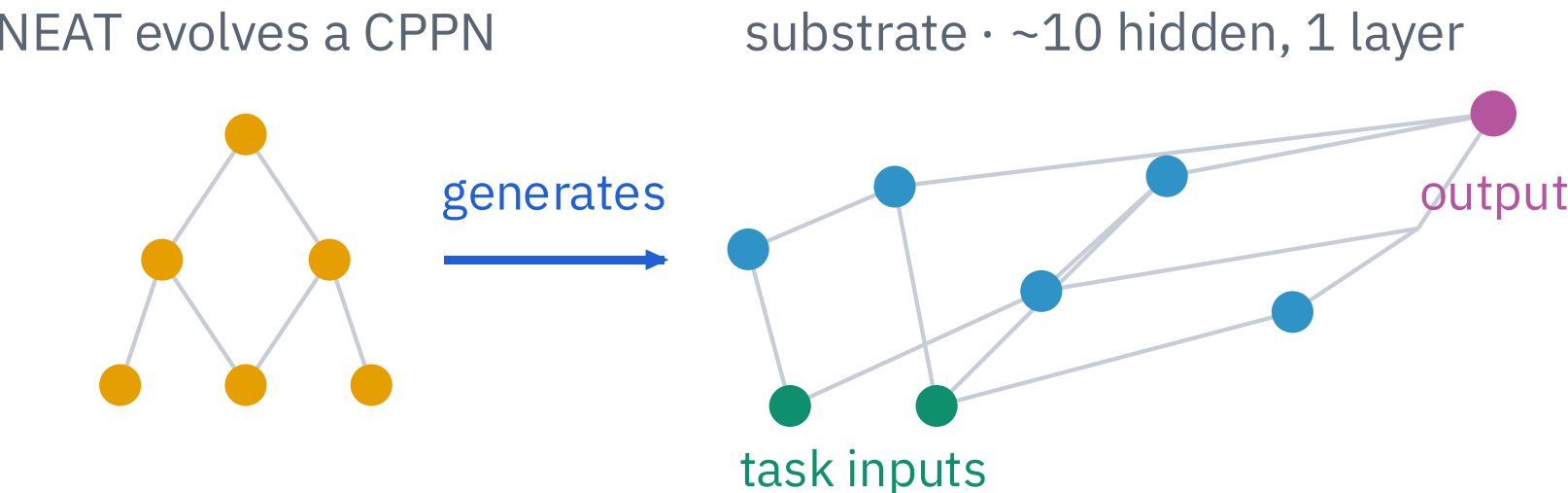
rare in practice: one activation everywhere its interaction with neuromodulation unexplored

RQ1 Does **activation function choice** create systematic multi-task barriers under neuromodulation? → 05

RQ2 Is the barrier **representational**, or **optimizer-dependent**? → 06 · A1

RQ3 Does **per-task activation selection** resolve it, and is neuromodulation still necessary? → 07–09

One small evolved substrate, five tasks at once



EMR-HyperNEAT (Tensor-Accelerated *ES-HyperNEAT*).

Indirect encoding, no backpropagation. Deliberately small: isolates the search question from capacity effects.

Fitness = **product of per-task accuracies** , nothing sacrificed. Converged $\Leftrightarrow$ all tasks $\geq 98\%$.

FIVE BOOLEAN TASKS · TWO COMPUTATIONAL CLASSES

	00	01	10	11	class
XOR	0	1	1	0	parity · 0→1→0
AND	0	0	0	1	threshold
OR	0	1	1	1	threshold
NAND	1	1	1	0	threshold · ¬AND
NOR	1	0	0	0	threshold · ¬OR

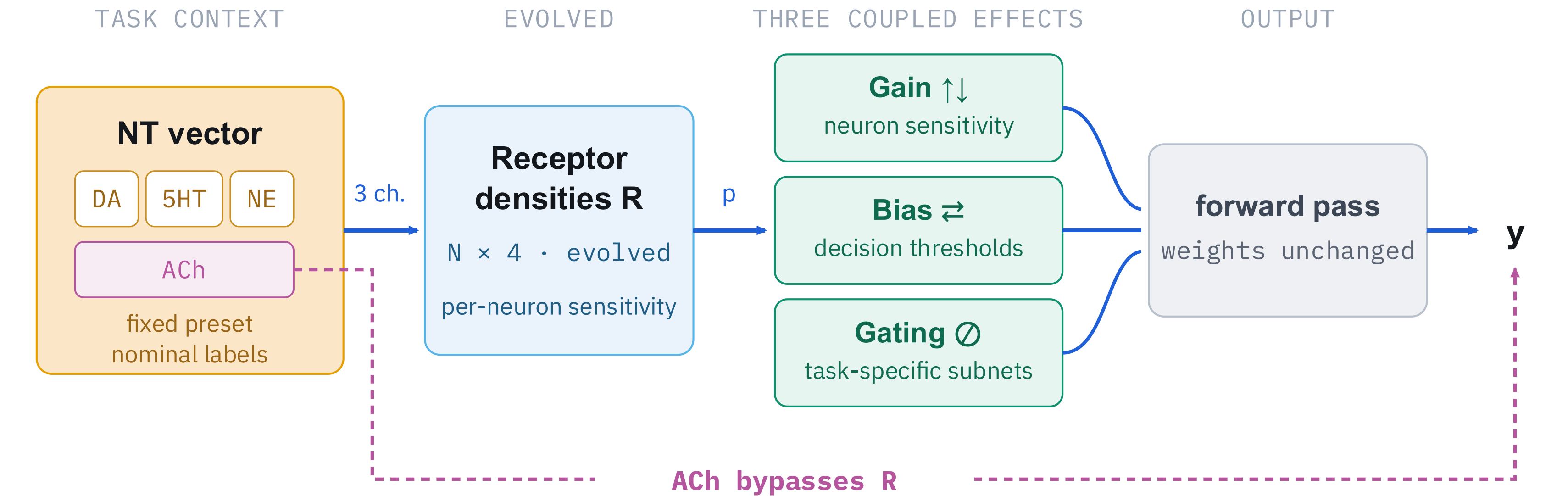
One cut for thresholds; XOR alternates 0→1→0.
NAND / NOR are complements of AND / OR, reused via polarity.

10,539 experiments

30 seeds / condition

JAX

Same weights, task-specific chemical context



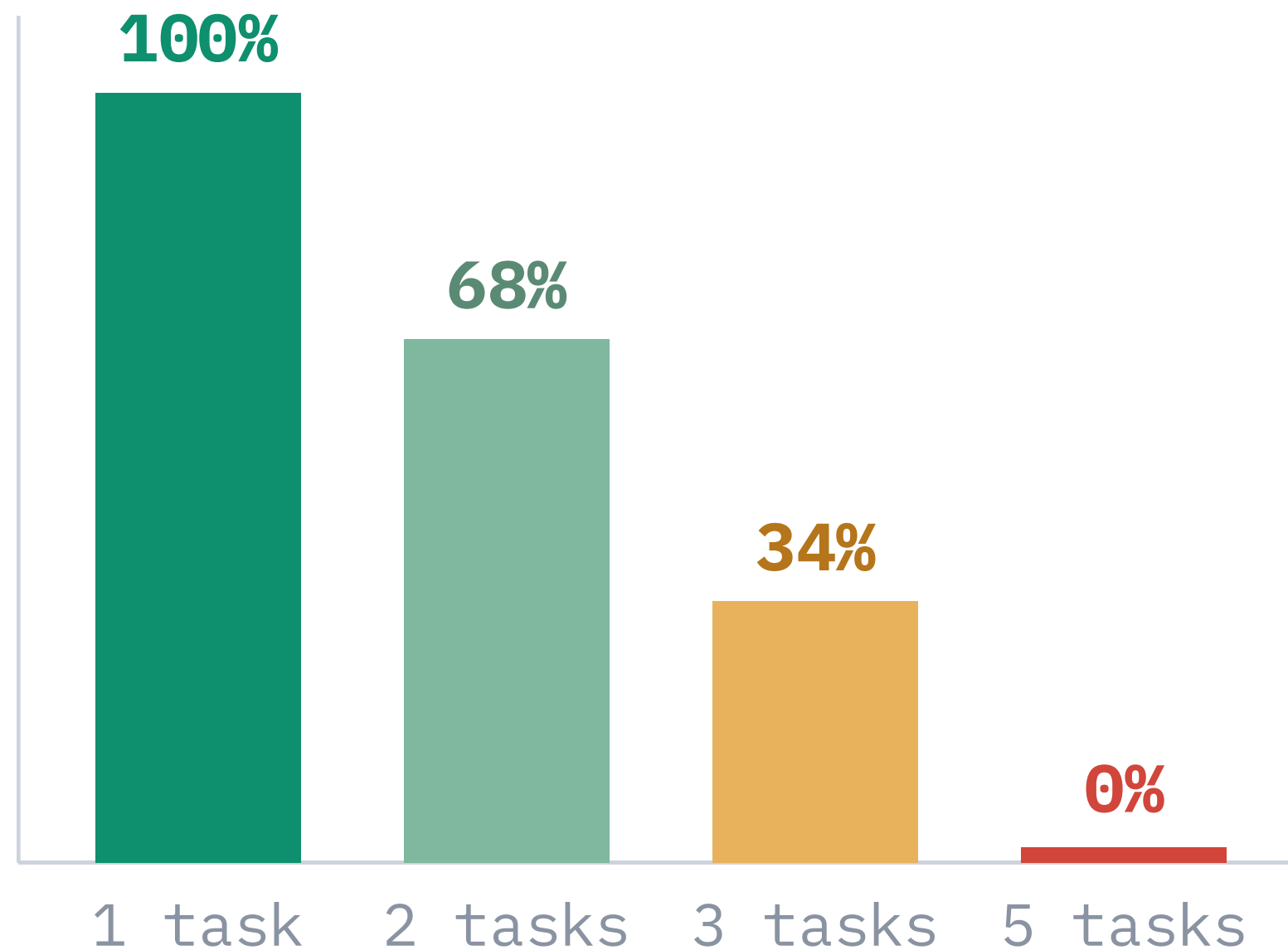
Post-pass, mechanical: if $ACh = 0$ the output is inverted, $y \leftarrow 1 - y$

NAND / NOR reuse the AND / OR profiles at no extra hidden-layer cost

XOR [.95 .05 .95 · 1]	AND [.10 .90 .10 · 1]	OR [.50 .50 .50 · 1]	NAND = AND, ACh 0	NOR = OR, ACh 0
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With one uniform activation, evolution hits a wall

CONVERGENCE RATE · UNIFORM TANH · 54 CONDITIONS



exactly 75%

XOR plateaus at 3 / 4 patterns with **zero variance**, every seed, while threshold tasks sit at 100%.

10×

budget (1,000 gens): still 0%, XOR pinned at 0.750

pair-specific

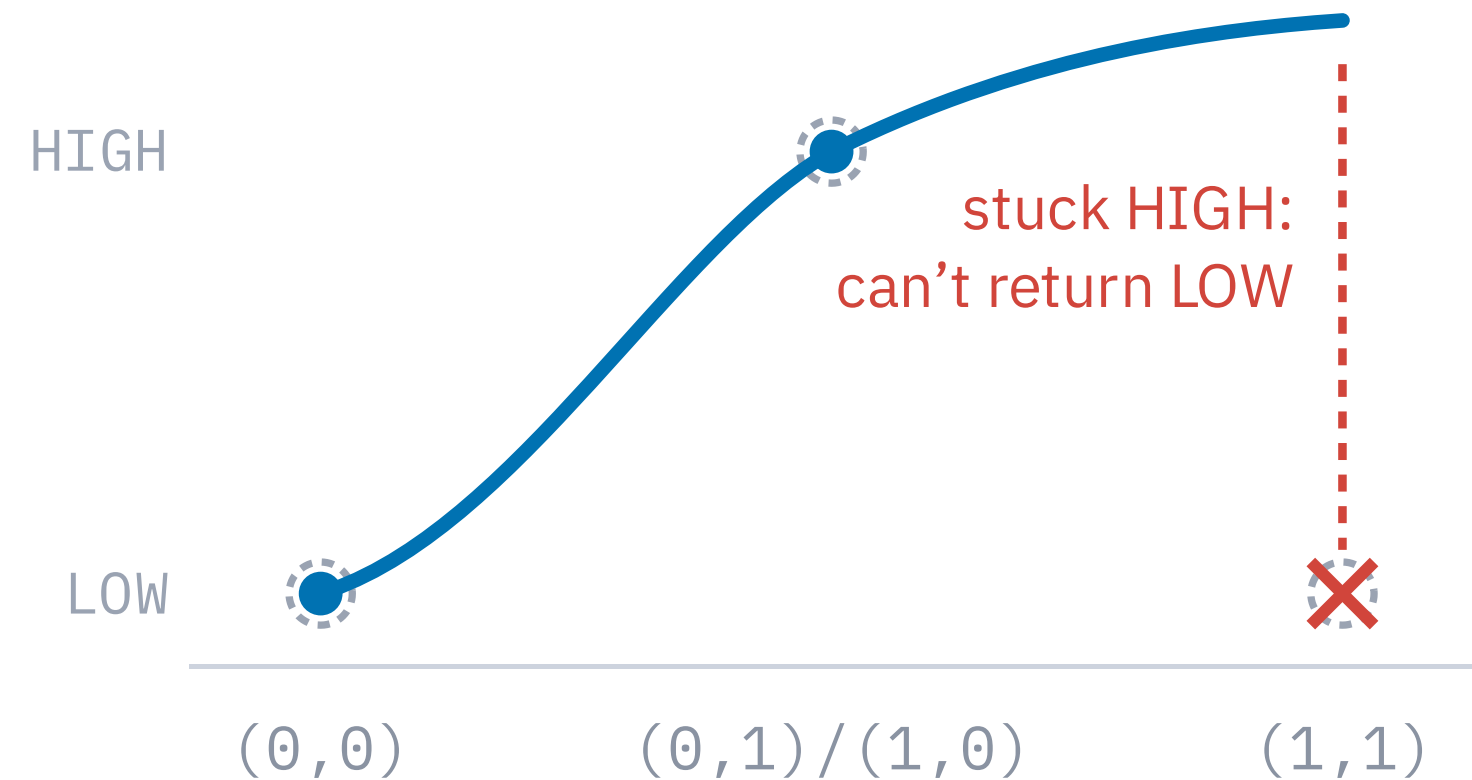
AND+OR: 100% ·
XOR+AND: 0%

A convergence barrier. More runtime does not help.

Monotonic functions can't turn around

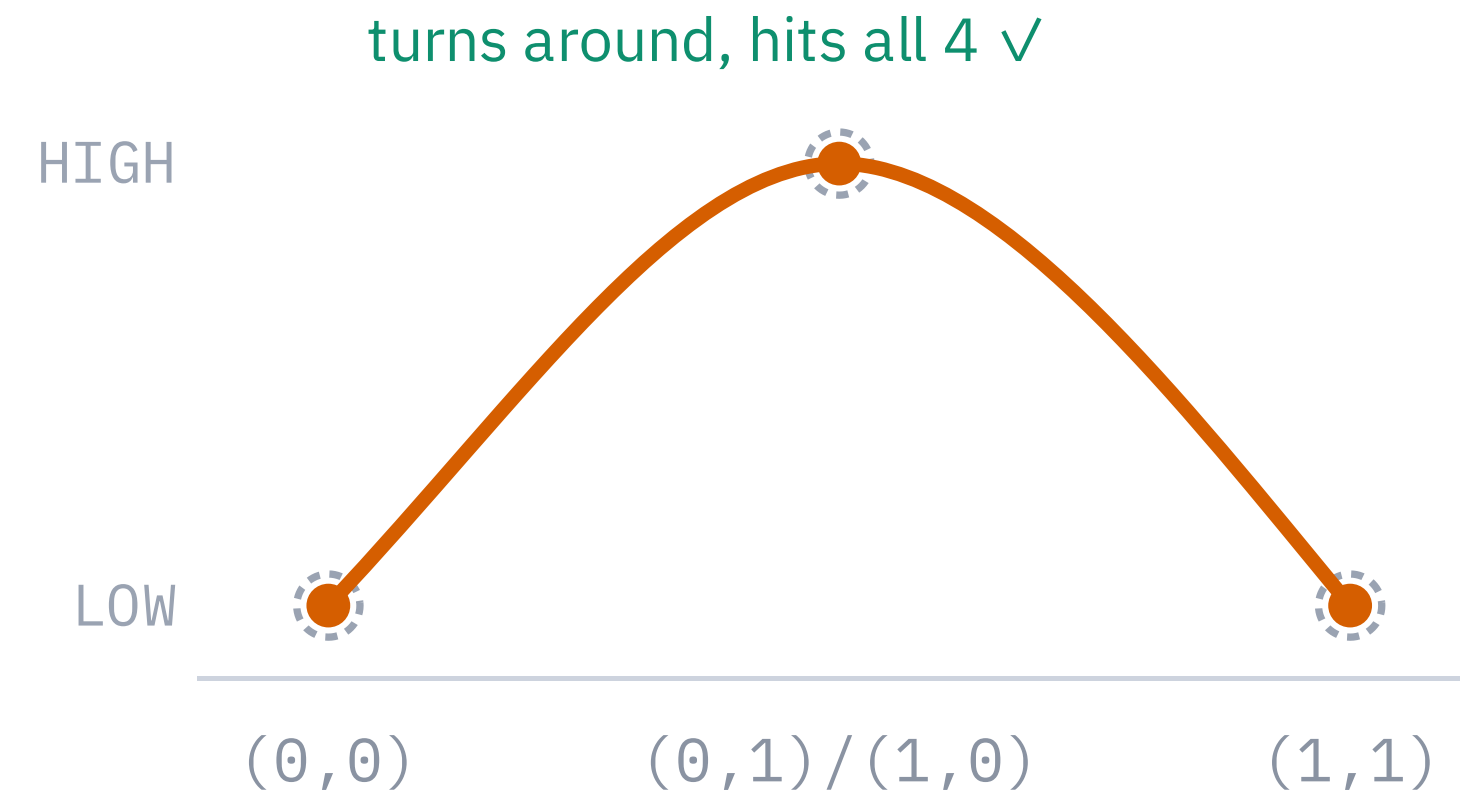
tanh · monotonic

3 / 4 → 75%



$\sin(\pi x/2)$ · oscillatory

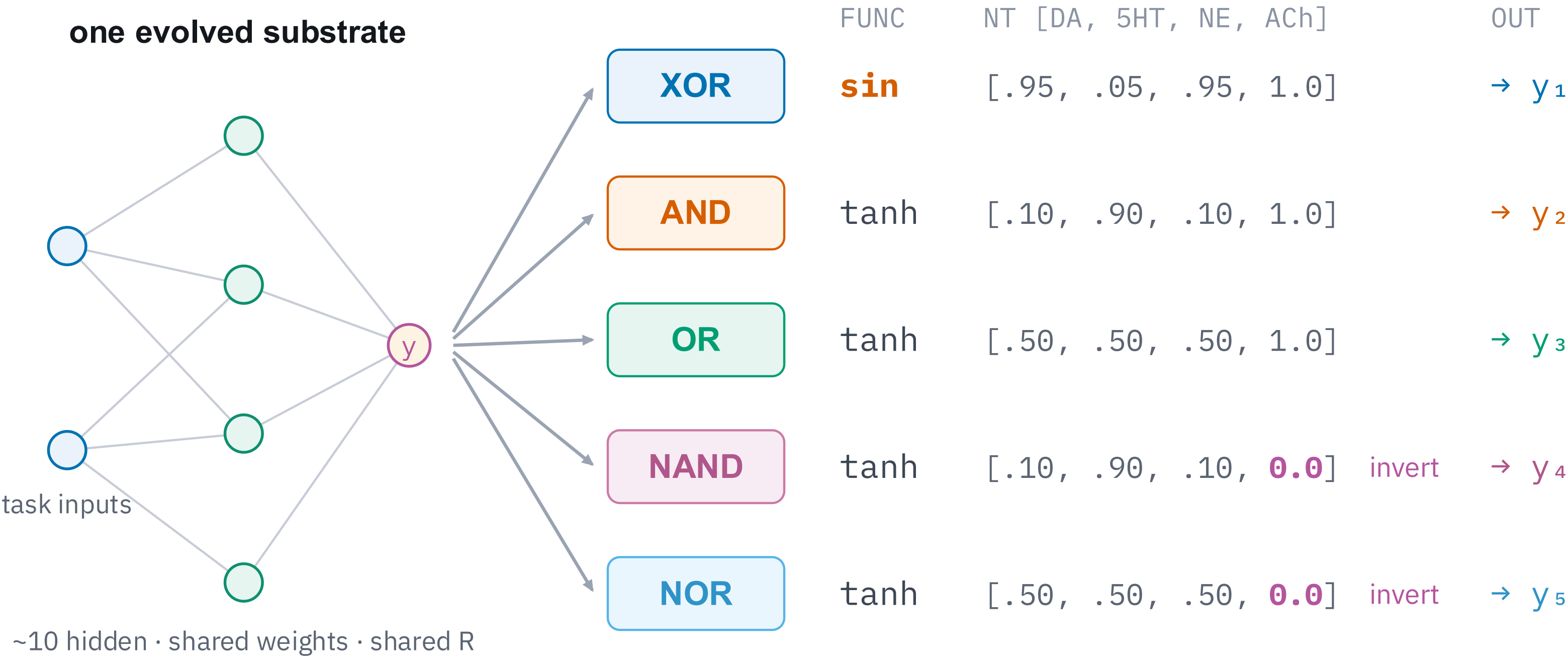
4 / 4 → 100%



XOR's required output is **LOW** → **HIGH** → **LOW** as the input sum grows (dashed circles). A monotonic curve can only keep climbing, so **every seed lands on the same 3 / 4 patterns: exactly 75%**.

Same architecture, uniform tanh: **100% under Adam** gradient descent. The wall is **evolutionary search**, not representation. → A1

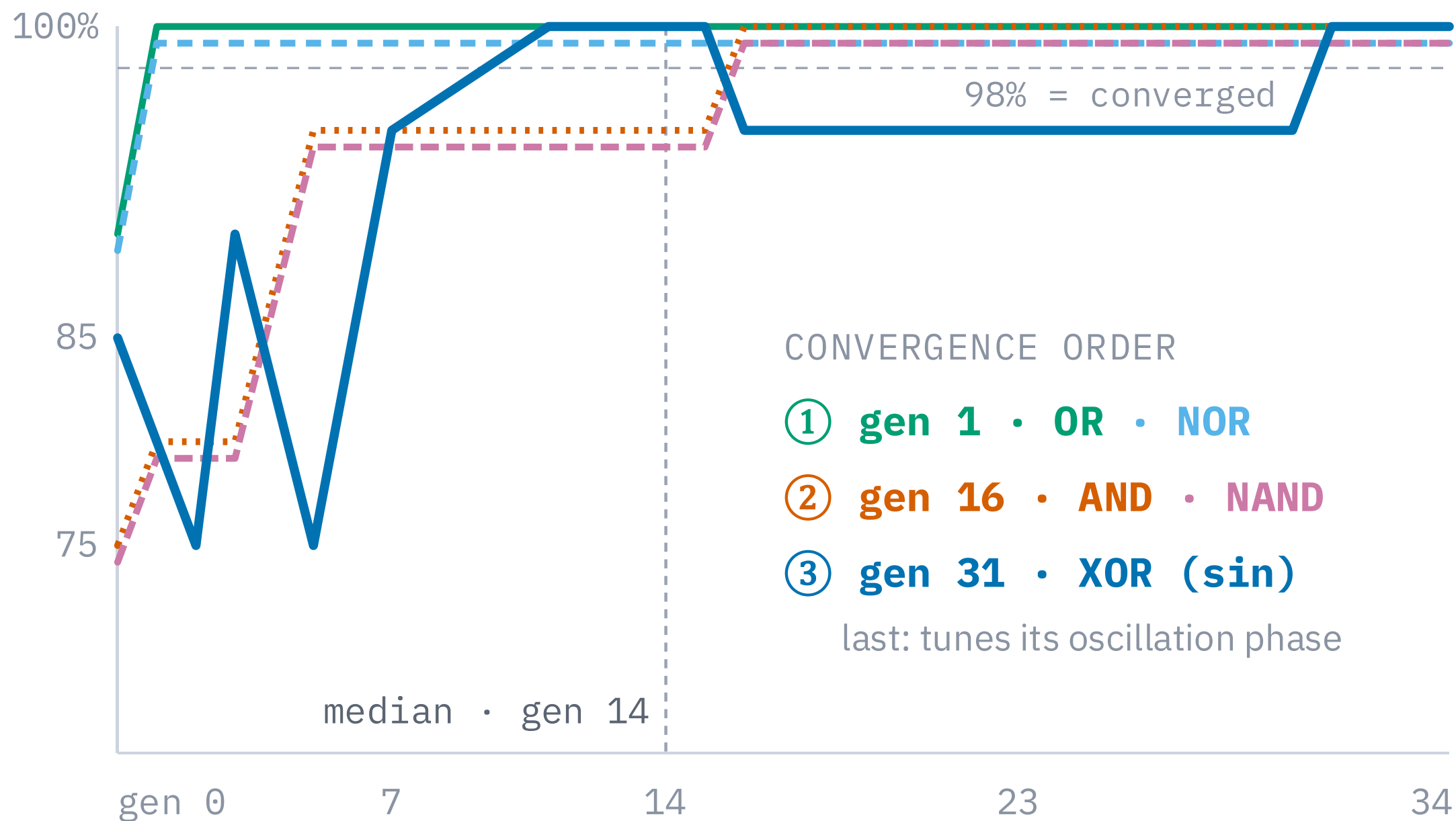
One network, five evaluation contexts



fitness = $y_1 \times y_2 \times y_3 \times y_4 \times y_5$ · all five at once, nothing sacrificed

THE RESULT

100% on all five tasks, every seed



30 / 30
seeds reach 100% on all 5 tasks

14 gens
median • 95% CI [12.7, 19.2] • range 4–36

VS. UNIFORM TANH

per-task	████████████████████	100%
tanh	████████████████████	0%

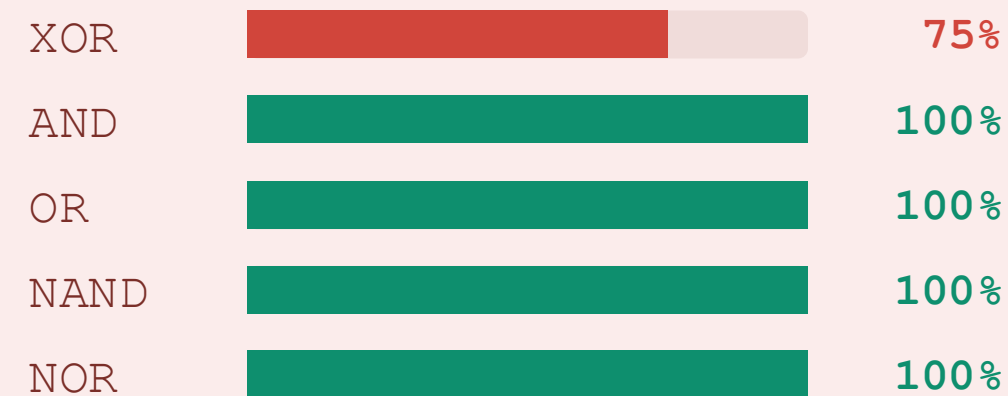
$p < 10^{-10}$ • Cramér's $V = 1.00$

Median accuracy • 30 seeds • per-task activation + neuromodulation

Neither mechanism suffices alone

– PER-TASK ACTIVATION (uniform tanh)

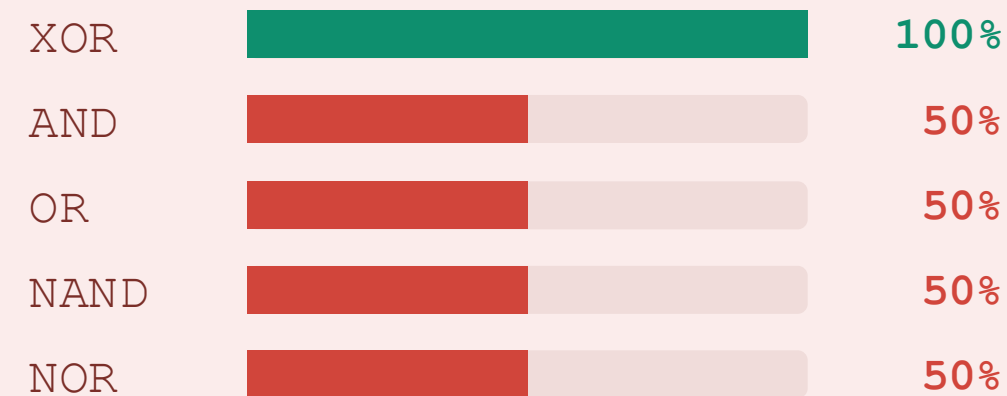
0% converged



the parity task is the casualty, pinned at 3 / 4 patterns

– TASK-SPECIFIC NT (flat modulation)

0% converged



XOR solves (sin suffices), thresholds stuck at chance

BOTH TOGETHER

100% 30 / 30 seeds



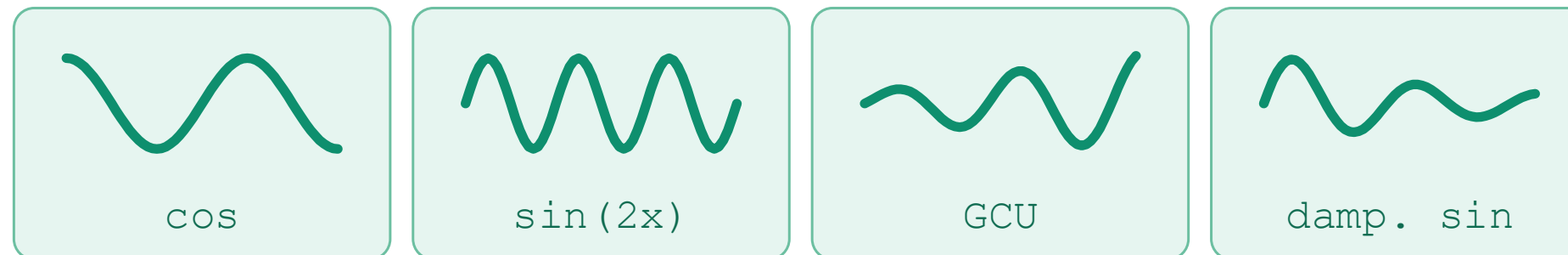
modulation separates thresholds; the primitive supplies the class

Control · uniform sin also converges: 30 / 30 , a statistical tie here (seed-matched, Wilcoxon $p = 0.88$). The barrier is **monotonicity** rather than single-function use, and sin-only degrades off this benchmark ($\rightarrow$ A15).

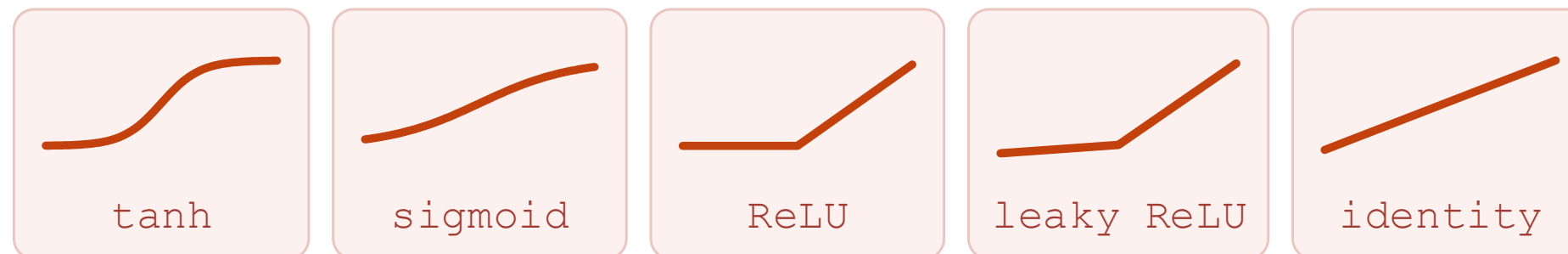
Also failed: per-task receptor matrices (5× search space, 50% fitness) · output blending (masks XOR at the 75% ceiling).

The mechanism, not a lucky configuration

OSCILLATORY → 100% ON ALL FIVE TASKS, EVERY FUNCTION



MONOTONIC → NEVER



every monotonic failure pins XOR at exactly 0.750

40×

modulation-strength range, every population size 100 → 1,000: **converges everywhere**

4× faster

than the 2-layer tanh alternative, with **276 vs 1,026** params

100%

at 3-input tasks, where uniform tanh drops to **26.7%**

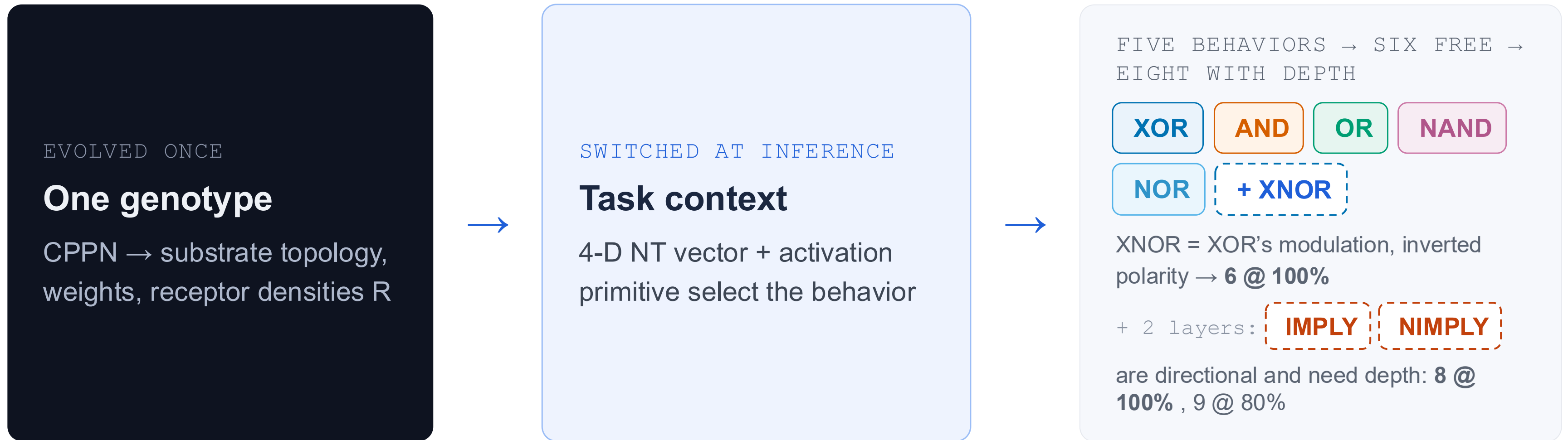
6 → 8

tasks at 100%: **XNOR free** via polarity, two layers carry the IMPLY family

Sin everywhere? Ties here, loses 11-0 on harder conditions (→ A15)

full robustness grid → A16

One genotype, many phenotypic behaviors



✗ retraining

✗ separate output heads

✗ per-task weights

✓ one 4-D signal + a primitive switch

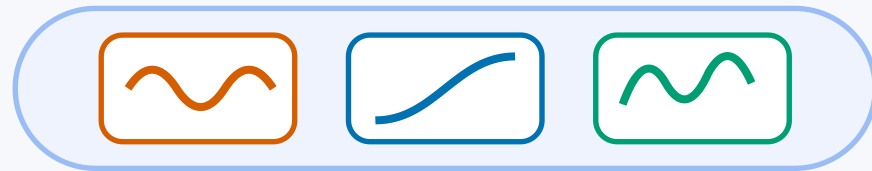
For science: environment-dependent phenotypic expression at the network level, what open-ended systems need when the task set isn't known in advance.

For engineering: one stored network, behaviors switched by a signal. Reconfiguration with **2.9× fewer parameters**, no retraining.

Let evolution own the primitive: three paths

1 · EVOLVED

Activation in the genotype

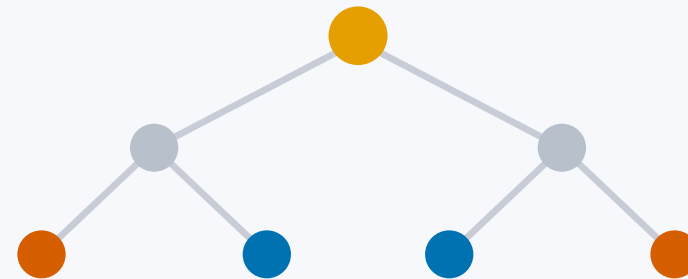


Make function selection heritable, so evolution meta-learns which primitives to keep in the palette.

The 2-task compatibility matrix maps **achievable behavioral combinations** : terrain for quality-diversity search.

2 · DEVELOPMENTAL

Differentiated neuron types

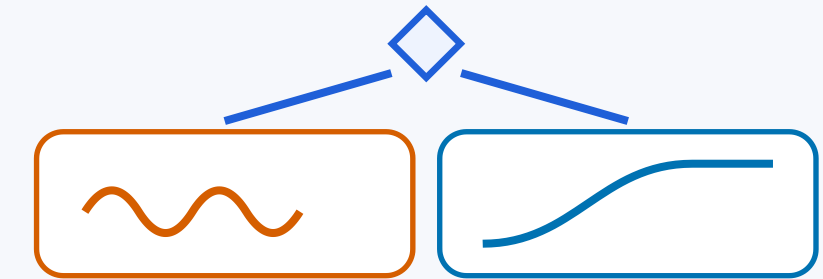


Evolve developmental rules that assign computational roles as the network forms.

Co-evolve the context: NT vectors evolved with the substrate; task identity inferred from the input stream.

3 · MODULAR

Composed specialists



Subnetworks with task-appropriate primitives, composed via routing: MoE at the primitive level.

Open question: freeze the substrate, search only NT space.
Can it bootstrap behaviors it was never evolved for? → A14



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THANK YOU

Questions?

Multi-Behavioral Evolved Substrates Through Neuromodulation and Activation Selection

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One genotype, five behaviors at inference

The barrier is evolutionary, not representational

Neuromodulation + primitive diversity, jointly necessary

`romain.claret@ucd.ie` · `github.com/RomainClaret/emr-hyperneat`



Three practice-changers for evolutionary multi-task systems

1

Audit the activation palette against the task set

A class mismatch shows up as a hard **search barrier** , invisible to single-task analysis.



2

Tasks span classes? Prefer oscillatory primitives

The advantage is **class-wide** : four oscillators, all 100%. Off the benchmark, **matching per task wins**.



3

Exploit output polarity

Complementary tasks come almost free through **post-pass inversion** at no extra hidden-layer cost.



The computational primitive is an **evolvable trait** rather than a fixed design choice.

Optimizer independence: the barrier is the search, not the net

SAME 2→10→1 NEUROMODULATED ARCHITECTURE · 5 TASKS

Adam · uniform tanh (lr .01) 100% · med 190 steps

Adam · per-task activation 100% · med 40 · 4.75× faster

Adam · uniform tanh (lr .001) 90%

(μ + λ)-ES · uniform tanh (direct) 76.7%

Frozen substrate (causal check)

tanh-trained weights + sin applied *post-hoc* : XOR at chance, **0 / 30 solved** . Retrain from the same weights with per-task activation: converges in **2 generations** .

The barrier operates **during training dynamics** , not at evaluation, and gradients navigate the coupled weight-receptor space that mutation-driven search did not.

2-task convergence rates (n = 30 per pair)

PAIR	INDEP. UNIFORM	COMP. UNIFORM	COMP. PER-TASK
XOR+NOR	100%	83.3%	100%
XOR+OR	86.7%	86.7%	100%
XOR+NAND	46.7%	0%	100%
XOR+AND	0%	0%	100%
AND+OR · AND+NAND · OR+NAND · NAND+NOR	100%	100%	100%
OR+NOR	0%	100%	100%
AND+NOR	16.7%	100%	100%

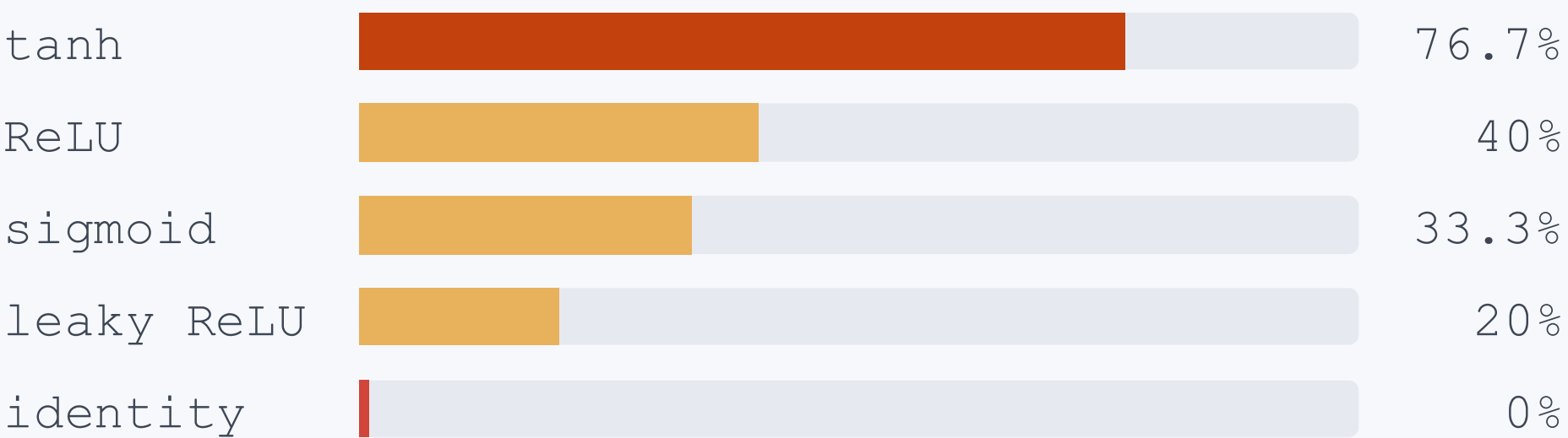
Independent: unique 4-D NT per task (ACh = 1.0). *Compositional* : NAND / NOR reuse AND / OR profiles with ACh = 0.0. **Per-task activation eliminates every barrier.**

The divide is the class, not the function

OSCILLATORY · 5-TASK · ALL 100%



MONOTONIC · NONE REACH 100%



Every monotonic failure

shows XOR at **exactly 0.750** : the same wall, five functions.

Arity-dependent at 3-input

cos 90% · GCU 90% · damp. 56.7% · sin(2x) 40%.
Doubled frequency $\Rightarrow$ rugged surface. Standard-period oscillators stay ahead (tanh: 26.7%).

Experiment inventory: 10,539 runs

BASELINE · 3,969

Benchmark: combos × topo × pop × seeds 1,665

Validation + ablations (per-task, 2-task, comp.) 2,304

ROBUSTNESS · 6,570 · 46 CONDITIONS

Task scaling + NT geometry 2,040

Domain generality (2D / 4D / Gaussian XOR) 1,200

Higher arity (3–5 input) · depth (1–5 layers) 960 + 780

Population & parameter sensitivity 930

Optimizer, oscillatory class, ablations, ext. gens 660

Grand total: 10,539 runs

JAX · CPU

30 seeds / condition

100–200 gens (2K for Adam)

Key results, statistically

FINDING	RATE	VS. CONTROL	p	$\phi^{\wedge}$
Per-task activation (1L, ES)	100%	0% tanh	$<10^{-10}$	1.00
Adam bypasses barrier (tanh)	100%	76.7% ES	0.011	0.36
Oscillatory class (4 fns)	100%	76.7% tanh	$<10^{-5}$	0.44
Capacity irrelevant (40n, tanh)	0%	0% (20n)	1.000	0.00
Neuromod necessary (flat NT)	0%	100% full	$<10^{-10}$	1.00
Depth resolves (tanh, 2L)	100%	76.7% (1L)	0.011	0.36
Frozen substrate (sin post-hoc)	0% XOR	100% fresh	$<10^{-10}$	1.00

Fisher's exact (2 groups) · χ^2 (multiple) · Cramér's V ($\phi^{\wedge}$) · $\alpha = 0.01$, Holm-Bonferroni · convergence = all tasks $\geq 98\%$

Where it scales, and where it stops

Arity: advantage persists, then narrows

3-input: per-task **100%** vs. tanh 26.7% (pop-ceiling 50%; Adam: 100%). 4-input: per-task 30% vs. 2L tanh 73.3%. 5-input: per-task **83.3%** vs. 2L 60%, with **276 vs. 1,026 params** .

Task count: a sharp, diagnosable cliff

6 tasks (XNOR shares XOR's modulation, inverted): **100%**. 7–10 tasks: 0% (1L indirect), and **every failure is the IMPLY family** . 2-layer: 8 tasks 100%, 9 at 80%, 10 at 30–50%.

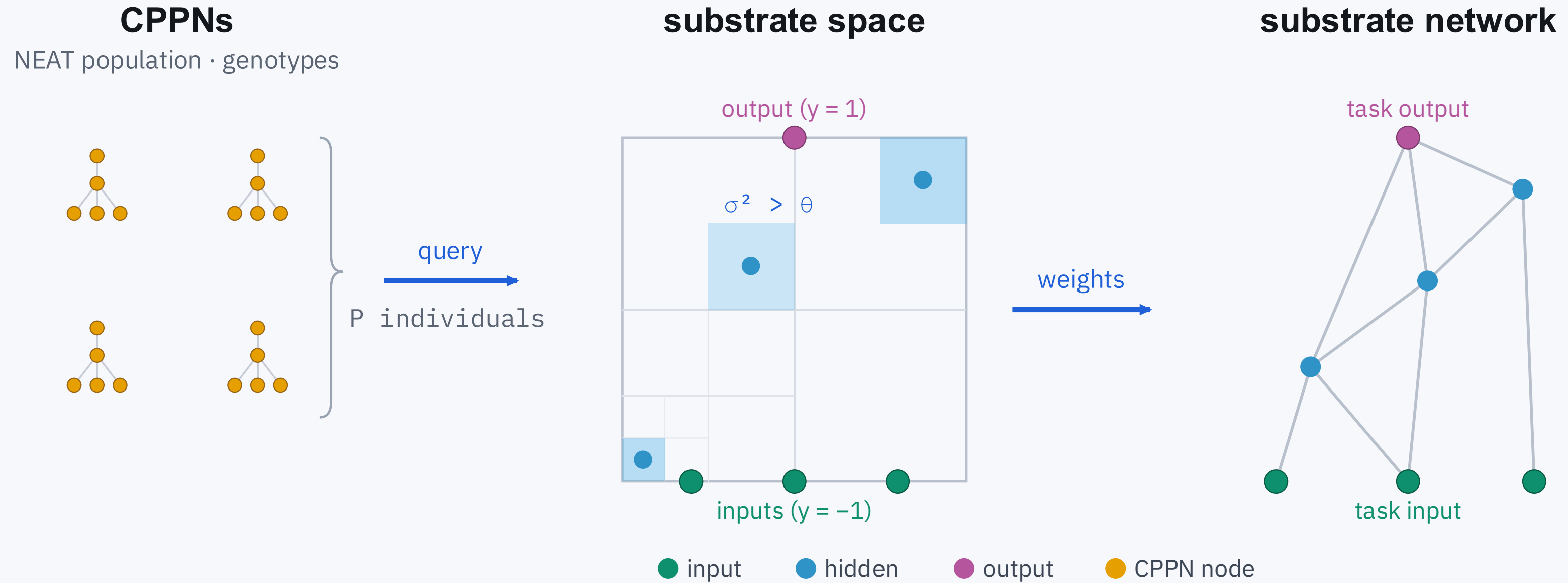
Capacity × depth

Doubling width (20→40n) at 1 layer: still 0%, **not capacity-limited** . With depth, capacity helps: 2L·40n reaches 76.7% at 4-input, Parity-4 solved in all seeds.

Domains & excess diversity

Continuous regression: no barrier (tanh 100%). Gaussian-XOR structure re-creates it (tanh 3.3% vs. per-task 50%). At 10 tasks, 10 unique activations: **0%** : diversity must match structure, not maximize variety.

EMR-HyperNEAT: from CPPN to substrate



A compact CPPN genome **generates** the whole network. EMR = eager multi-resolution, tensor-accelerated (JAX).

Limitations

Task complexity

Boolean tasks are clean but simple; per-task degrades at 4-input (30%), and continuous domains without parity structure show no barrier.

Hand-designed NT presets

Vectors were specified, not searched; 1 / 10 random sets works (polarity preserved). Co-evolving NT with the substrate is the next step.

Explicit task identity

The task is announced at test time; biological modulation infers context, so an artificial analog should read it from the input stream.

Encoding & optimizer scope

Indirect vs. direct barrier strengths differ (0% vs. 76.7%), illustrative rather than controlled. ($\mu+\lambda$)-ES only; QD / novelty search untested.

In embodied settings, an agent could transform the task by acting (e.g. reorienting its body). This enabler is specific to the disembodied Boolean setting studied here.

The mechanism, precisely

1 · PROJECTION

$$\mathbf{p} = \mathbf{R}_{:,1:3} \cdot \mathbf{NT}_{1:3}$$

$\mathbf{R} \in \mathbb{R}^{N \times 4}$ evolved receptor densities:
per-neuron sensitivity to DA, 5HT,
NE

2 · MODULATED FORWARD PASS

$$\mathbf{h}_{\text{out}} = \mathbf{f}(\mathbf{g}_{\text{eff}} \odot \mathbf{h} + \mathbf{b}_{\text{mod}}) \odot \boldsymbol{\gamma}$$

$$\mathbf{g}_{\text{eff}} = \mathbf{g}_{\text{base}} + s \cdot \mathbf{p}$$

$$\mathbf{b}_{\text{mod}} = s \cdot \mathbf{p}$$

$$\boldsymbol{\gamma} = \sigma(\mathbf{p})$$

3 · POLARITY (ACh) · POST-PASS

$$\mathbf{y}_{\text{final}} = \mathbf{ACh} \cdot \mathbf{y} + (1 - \mathbf{ACh})(1 - \mathbf{y})$$

$\mathbf{ACh} \in \{0, 1\} \rightarrow$ identity or inversion,
applied **after** the forward pass;
NAND/NOR from AND/OR weights

Gain, bias, and gating all derive from the **single projection $\mathbf{p}$** , coupled by design. Nuance: $\mathbf{R}$ is declared **$N \times 4$** but the pass reads only columns 1–3, so **ACh never routes through $\mathbf{R}$** ; its 4th column is vestigial. Letting evolution couple to it $\rightarrow$ co-evolution future work (A14).

$$s = 5.0 \text{ (fixed)}$$

$$\text{gains clipped } [0.1, 5.0]$$

$$\mathbf{y} = \sigma(\mathbf{h}_{\text{out}}^T \mathbf{W}_{\text{out}})$$

“Oscillatory solves parity is known, so what’s new?”

KNOWN · SINGLE-TASK

GCU: single-neuron XOR solutions. **SIREN:** periodic activations preserve high-frequency structure. **NEAT lineage:** per-node activation evolution *within* a task.

THIS PAPER · THE INTERACTION

Under **neuromodulation + evolutionary search**, activation choice creates an **emergent multi-task barrier**: pair-specific (AND+OR 100%, XOR+AND 0%), optimizer-dependent, invisible to single-task analysis. Mapped across **10,539 runs** with causal tests (optimizer independence, frozen substrate).

MECHANISM PLACEMENT

Not per-node diversity: **one function for all neurons, switched per task**, so the whole substrate changes computational mode at inference. MoE routes inputs to experts; this routes the **primitive itself**.

Related threads: neural reuse keeps components multifunctional by re-composing them; here the network stays fixed and only primitive + modulatory context change. Neuromodulation lineage (Soltoggio → Ellefsen → Backpropamine) modulates existing computations. It cannot make a monotonic function oscillate.

Why these design choices?

Why product fitness, not mean?

$F = \prod a_t$ enforces **simultaneous mastery**. A mean lets evolution sacrifice XOR and still score 0.8. Output blending did exactly that, settling at the 75% ceiling.

Is the protocol the barrier? No.

Sequential evaluation (one task at a time, switching NT) also succeeds under per-task activation. The conflict is activation–task mismatch, not simultaneous evaluation.

Convergence & statistics

Converged $\Leftrightarrow$ **all tasks $\geq 98\%$** ; metric = convergence rate over **30 seeds**. Fisher's exact / χ^2 , Cramér's V, $\alpha = 0.01$ with Holm-Bonferroni.

NT presets: designed, not searched

Hand-placed to separate threshold tasks in modulatory space. Random NT: **1 / 10 sets reaches 100%** when ACh polarity is preserved. Co-evolution is future work.

Which modulation component does the work?

2-INPUT · REDUNDANT

Gain-only, bias-only, gating-only:
each 100% alone , but 1.7× slower
than the coupled formulation.

3-INPUT · REDUNDANCY BREAKS

Bias-only: **100%** · gain-only: **63.3%**
. Bias provides the offsets that
separate 8-pattern truth tables at
higher arity.

POLARITY-ONLY · FAILS

Flat DA/5HT/NE with correct ACh:
0%, stuck at the 75% ceiling .
Inversion alone can't separate
thresholds.

Three distinct jobs: **polarity** handles complements · **gain / bias / gating** separates thresholds · **the activation primitive** supplies the computational class.

Recurrence? Continuous domains?

TOPOLOGY · ARITY-DEPENDENT

2-input: all **six topologies identical** (feedforward → fully recurrent). 3-input, uniform tanh: feedforward **20%** vs. fully recurrent **80%** . Per-task activation saturates it: **both 100%** : the primitive dominates the topology.

DOMAINS · STRUCTURE DECIDES, NOT TYPE

Cont. regression	tanh 100% / p-t 43.3% / sin 6.7%
Gaussian XOR (parity)	tanh 3.3% / p-t 50% / 2L 93.3%
4-class Gaussian XOR	tanh 30% / per-task 100%
4-D Iris-like synthetic	only 2-layer: 80%

The activation–task interaction follows **computational structure** (parity-class vs. threshold), not Boolean vs. continuous.

How the threads combine

THIS PAPER · TIME

Per-task switching

One function for **all neurons** , switched **between tasks** , so the substrate changes computational mode at inference. NT presets fixed.

COMPANION WORK · SPACE

Per-node evolution

Each neuron evolves **its own function** : diversity **within one task** . Orthogonal axis: spatial, not temporal. Combined: evolve *which neuron runs which function under which task* .

THIRD THREAD · CONTEXT

Co-evolved NT vectors

Replace hand-designed presets: **1 / 10 random sets already works** (polarity preserved), a navigable space, and the compatibility geometry (conflict, distance, direction, polarity) structures its landscape.

UNIFIED →

A richer neuromodulation that selects the **activation function itself** , not only gain and bias. One co-evolved modulatory signal deciding **how much** a neuron computes and **what** it computes, per node, per task.

“Why not sin everywhere?” A tie here, not beyond

HEADLINE BENCHMARK · 2-INPUT

Both **30 / 30** · medians 14 vs. 17 · seed-matched paired Δ **-0.1 gen**, Wilcoxon **p = 0.88** : a statistical tie. No speed claim.

15 MATCHED CONDITIONS · 900 RUNS · SAME SEEDS

Per-task wins **11 · ties 4 · loses 0** · McNemar **75 vs. 5** ($p < 10^{-15}$) · pooled **42.4% vs. 26.9%**

THE MARGIN IS WIDEST WHERE TASK SETS MIX COMPUTATIONAL CLASSES

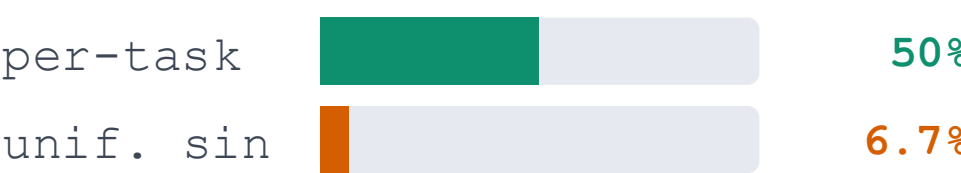
Parity-3 multi-task



5-task · 3-input



Gaussian XOR



Matching activations to task structure buys **robustness rather than speed** . Sufficiency on one benchmark doesn't make sin the safer default. (And on continuous regression sin drops to 6.7% while tanh takes it outright → A13.)

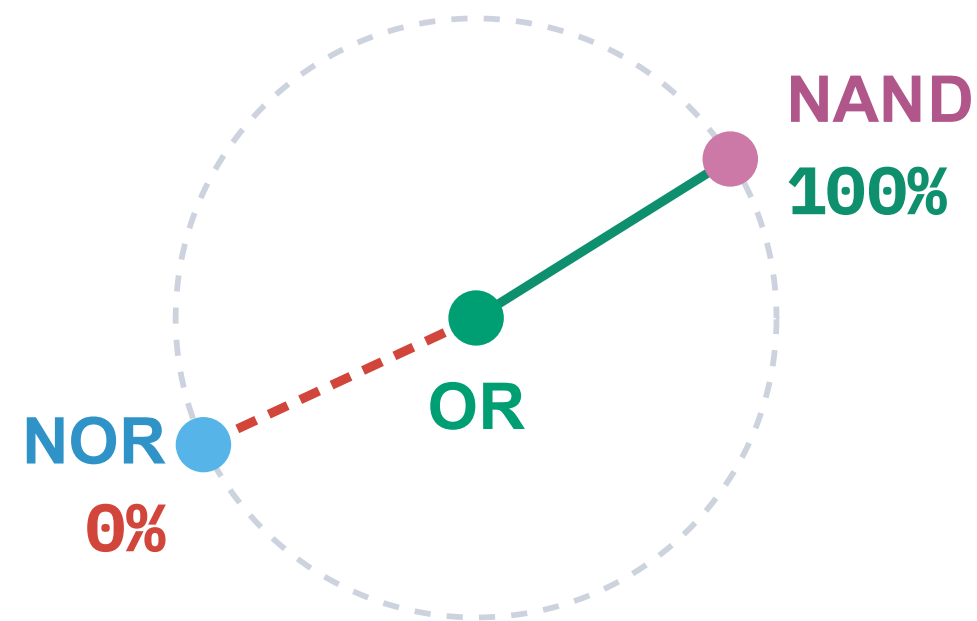
Robustness at one glance

WHAT WE VARIED	WHAT HAPPENED	OUTCOME
ACTIVATION CLASS	cos, sin(2x), GCU, dampened sin all solve the five tasks. No monotonic function does: every failure pins XOR at 0.750.	4 / 4 @ 100% monotonic: none
HYPERPARAMETERS	Modulation strength 0.5× → 20× (a 40× range) and every population size 100 → 1,000 : converges everywhere.	100% no tuning needed
DEPTH ALTERNATIVE	Two-layer uniform tanh also reaches 100%. Per-task at one layer gets there 4× faster with 276 vs 1,026 params.	4× faster 2.9× leaner
HIGHER ARITY	3-input tasks: per-task activation holds 100% ; uniform tanh drops to 26.7% , Parity-4 under tanh hits 0%.	100% tanh: 26.7% → 0%
TASK COUNT	XNOR joins free via inverted polarity → 6 @ 100%. One-layer cliff at 7, all failures IMPLY ; two layers hold 8.	6 → 8 @ 100% 9 @ 80% (→ A6)
UNIFORM SIN INSTEAD?	Sin for every task also solves all five here. On 15 harder matched conditions it loses 11 and wins none against per-task.	per-task: 11–0 42.4% vs 26.9% (→ A15)

Asymmetric tasks (IMPLY) hit the same wall: symmetric gain/bias/gating can't encode direction; sin resolves it (→ A13).

Distance alone doesn't predict coexistence

NT SPACE · SAME RADIUS FROM OR

same distance · $d = 0.566$ 

OR + NAND

100%

00	01	10	11
X	✓	✓	X

truth tables disagree on 2 / 4 patterns

OR + NOR

0%

00	01	10	11
X	X	X	X

complements: disagree on 4 / 4 patterns

Same modulatory distance, opposite fates. Distance alone: $R^2 = 0.31$ (20 pair × schema observations).

THREE INDEPENDENT RESCUES FOR OR + NOR

flip polarity (ACh)

→ 100% at generation 0

reorient (DA→NE) · same d

→ 100% at generation 1

more distance

→ AND+NAND at $d = 1.20$: 100%

Under per-task activation **all of these patterns vanish**: artifacts of activation–task mismatch, never fundamental.